The PWDG–UWVF Equivalence Revisited

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Abstract

The ultra-weak variational formulation (UWVF) is known to be the Trefftz discontinuous Galerkin scheme obtained from the standard PWDG flux family by setting $\alpha = \beta = \delta = 1/2$. We give a short structural explanation of this fact. On every interior mesh edge, the two weighted PWDG jump contributions at these parameter values equal one half of the sum of the squared norms of the two oriented impedance-transmission defects used by the UWVF. Thus the Trefftz-DG norm is the natural Riesz norm of the global transmission defect. In PWDG coordinates the skew-Hermitian part of the discrete matrix is iG, where G is the TDG Gramian, and Riesz normalization puts the operator in the normal form $K + iI$, with K Hermitian. The same argument applied directly to the discrete UWVF yields the complementary normal form $\frac{1}{2}I + iK_{\mathrm{U}}$. This identifies the classical equivalence as a relation between broken-field and impedance-trace coordinates for the same edge-coupling problem, rather than only as a particular choice of penalties.

Keywords: Trefftz methods, plane-wave discontinuous Galerkin, ultra-weak variational formulation, impedance traces, Riesz map, conditioning

1. Introduction

The ultra-weak variational formulation (UWVF) of Cessenat and Després uses impedance traces on element boundaries as unknowns [1, 2]. Plane-wave discontinuous Galerkin (PWDG), or more generally Trefftz-DG, instead uses broken Helmholtz fields and couples neighboring elements by numerical fluxes. The two methods have long been known to coincide for the particular choice

$$\alpha = \beta = \delta = \frac{1}{2}; \tag{1}$$

see [3, Remark 2.1] and [6, Sec. 2.2]. The usual derivation verifies this by inserting the UWVF fluxes into the DG formulation. It does not, however, make transparent why the factors 1/2 arise.

The point of this note is that (1) is forced by a simple impedance identity. The Dirichlet and normal-flux jumps used by PWDG are the two real channels of the complex transmission residual used by the UWVF. At the values (1), their weighted squares are exactly the norm of the two oriented impedance defects on each mesh edge. This also gives a direct Riesz interpretation of the Trefftz-DG norm. Since

$$\operatorname{Im} A_h(v, v) = \|v\|_{\mathrm{TDG}}^2 \tag{2}$$

for Trefftz fields [6], the skew-Hermitian part of the PWDG matrix is iG, where G is the corresponding Gram matrix. Its inverse square root therefore produces a normal form canonical relative to the TDG Riesz metric, rather than merely rescaling an ill-conditioned plane-wave basis. The stability identity itself and the ensuing matrix algebra are standard; the new observations here are the edgewise impedance isometry explaining (1) and the paired PWDG–UWVF Riesz normal forms.

For direct comparison with the classical UWVF literature we use the sign convention $\partial_n u + iku = g$. Conjugating all impedance traces gives the corresponding formulas for the opposite time convention.

2. The two formulations

Let $\Omega \subset \mathbb{R}^2$ be polygonal and let $\mathcal{T}_h$ be a mesh with interior edges $\mathcal{F}_h^I$ and boundary $\Gamma = \partial\Omega$. We consider

$$-\Delta u - k^2 u = 0 \quad \text{in } \Omega, \qquad \partial_n u + iku = g \quad \text{on } \Gamma. \tag{3}$$

For an interior edge $e = \partial K^+ \cap \partial K^-$, with outward normals $n^\pm$, set

$$[\![v]\!]_N = v^+ n^+ + v^- n^-, \qquad [\![\nabla v]\!]_N = \nabla v^+ \cdot n^+ + \nabla v^- \cdot n^-,$$

and use the usual arithmetic averages $\{\{v\}\}$ and $\{\{\nabla v\}\}$. For each K, let $V(K) \subset \{v \in H^2(K) : -\Delta v - k^2 v = 0\}$ be a finite-dimensional local Trefftz space (typically plane waves), and set

$$V_h := \{v \in L^2(\Omega) : v|_K \in V(K)\ \forall K \in \mathcal{T}_h\}.$$

2.1. PWDG

The standard primal Trefftz-DG formulation [3, 6] is: find $u_h \in V_h$ such that

$$A_h(u_h, v_h) = \ell_h(v_h) \qquad \forall v_h \in V_h, \tag{4}$$

where, for $\alpha, \beta > 0$ and $0 < \delta < 1$,

$$\begin{aligned} A_h(u,v) = \int_{\mathcal{F}_h^I} \Big(&\{\{u\}\}\, [\![\nabla \bar{v}]\!]_N - \{\{\nabla u\}\} \cdot [\![\bar{v}]\!]_N + \alpha ik\, [\![u]\!]_N \cdot [\![\bar{v}]\!]_N \\ &- \beta (ik)^{-1} [\![\nabla u]\!]_N [\![\nabla \bar{v}]\!]_N \Big)\, ds \\ + \int_\Gamma \Big(&(1-\delta) iku\bar{v} + (1-\delta) u\, \overline{\partial_n v} - \delta\, \partial_n u\, \bar{v} \\ &- \delta (ik)^{-1} \partial_n u\, \overline{\partial_n v} \Big)\, ds, \end{aligned} \tag{5}$$

and

$$\ell_h(v) = \int_\Gamma g \Big((1-\delta)\bar{v} - \delta (ik)^{-1} \overline{\partial_n v} \Big)\, ds. \tag{6}$$

The associated Trefftz-DG norm is

$$\begin{aligned} \|v\|_{\mathrm{TDG}}^2 = &k^{-1} \|\beta^{1/2} [\![\nabla v]\!]_N\|_{\mathcal{F}_h^I}^2 + k \|\alpha^{1/2} [\![v]\!]_N\|_{\mathcal{F}_h^I}^2 \\ &+ k^{-1} \|\delta^{1/2} \partial_n v\|_\Gamma^2 + k \|(1-\delta)^{1/2} v\|_\Gamma^2. \end{aligned} \tag{7}$$

For Trefftz functions, (2) follows directly from (5). For the impedance problem (3), (7) is a norm on the discrete Trefftz space: zero jumps and zero boundary contribution give a global homogeneous solution, hence zero by uniqueness.

2.2. UWVF

Let

$$\mathcal{X} = \prod_{K\in\mathcal{T}_h} L^2(\partial K).$$

For $y_K \in L^2(\partial K)$, let e_K solve the local adjoint-impedance problem

$$-\Delta e_K - k^2 e_K = 0 \text{ in } K, \qquad (-\partial_{n_K} + ik)e_K = y_K \text{ on } \partial K,$$

and define the local trace map

$$F_K y_K = (\partial_{n_K} + ik)e_K. \tag{8}$$

For the unit-impedance boundary condition in (3), the UWVF reads [1, 6]: find $x \in \mathcal{X}_h \subset \mathcal{X}$ such that, for all $y \in \mathcal{X}_h$,

$$\sum_{K\in\mathcal{T}_h} \int_{\partial K} x_K \bar{y}_K \, ds - \sum_{K\in\mathcal{T}_h} \sum_{K'\sim K} \int_{\partial K\cap\partial K'} x_{K'} \, \overline{F_K y_K} \, ds \\ = \sum_{K\in\mathcal{T}_h} \int_{\partial K\cap\Gamma} g \, \overline{F_K y_K} \, ds. \tag{9}$$

Here $K' \sim K$ denotes the element across an interior edge and

$$\mathcal{X}_h = \{x \in \mathcal{X} : x_K = (-\partial_{n_K} + ik)v_h|_K, \ v_h \in V_h\}.$$

Following Cessenat and Després, $x_K = (-\partial_{n_K} + ik)v_h|_K$ is outgoing and $F_K x_K = (\partial_{n_K} + ik)v_h|_K$ incoming [1, Def. 1.5]. The field is recovered from x_K. For $\alpha = \beta = \delta = 1/2$, (4)–(6) is exactly (9) in broken-field variables [3, Remark 2.1].

3. Why the parameters are one half

Consider one interior edge $e = \partial K \cap \partial K'$, and abbreviate

$$u_K = u|_K, \quad u_{K'} = u|_{K'}, \quad q_K = \partial_{n_K} u_K, \quad q_{K'} = \partial_{n_{K'}} u_{K'}.$$

Introduce the normalized outgoing and incoming traces

$$\gamma_K^- u = \frac{-q_K + iku_K}{\sqrt{2k}}, \qquad \gamma_K^+ u = \frac{q_K + iku_K}{\sqrt{2k}}. \tag{10}$$

Here γ_K^- is outgoing and γ_K^+ incoming [1, Def. 1.5]; the factor $\sqrt{2k}$ is only a convenient normalization and cancels from the Riesz-normalized UWVF matrix. Exact transmission is $\gamma_K^- u = \gamma_{K'}^+ u$ and $\gamma_{K'}^- u = \gamma_K^+ u$. Define the two oriented defects

$$r_K = \gamma_K^- u - \gamma_{K'}^+ u, \qquad r_{K'} = \gamma_{K'}^- u - \gamma_K^+ u.$$

Proposition 1 (Impedance identity)**.** *For every broken field with well-defined traces on e,*

$$\frac{1}{2}\big(\|r_K\|_e^2 + \|r_{K'}\|_e^2\big) = \frac{1}{2k}\|q_K + q_{K'}\|_e^2 + \frac{k}{2}\|u_K - u_{K'}\|_e^2. \tag{11}$$

Thus the interior part of (7) *for $\alpha = \beta = 1/2$ is exactly one half of the squared UWVF transmission defect.*

Proof. From (10),

$$r_K = \frac{-(q_K + q_{K'}) + ik(u_K - u_{K'})}{\sqrt{2k}}, \quad r_{K'} = \frac{-(q_K + q_{K'}) - ik(u_K - u_{K'})}{\sqrt{2k}}.$$

Use $|a+ib|^2 + |a-ib|^2 = 2|a|^2 + 2|b|^2$, valid for complex a, b, and integrate over e. $\square$

The boundary factor is equally explicit. For either element-boundary trace in (10),

$$\frac{1}{2}\left(\|\gamma^- v\|_\Gamma^2 + \|\gamma^+ v\|_\Gamma^2\right) = \frac{1}{2k}\|\partial_n v\|_\Gamma^2 + \frac{k}{2}\|v\|_\Gamma^2, \tag{12}$$

which is exactly the boundary contribution in (7) for $\delta = 1/2$. More generally, replacing (10) by $-\sqrt{\beta/k}\, q_K + i\sqrt{\alpha k}\, u_K$ and its outgoing counterpart produces the jump weights β/k and αk; thus α/β selects the effective impedance $k\sqrt{\alpha/\beta}$.

4. Riesz normal forms

Let $\{\Phi_j\}_{j=1}^N$ be a basis of V_h, and define the PWDG matrix and DG Gram matrix by

$$A_{j\ell} = A_h(\Phi_\ell, \Phi_j), \qquad G_{j\ell} = (\Phi_\ell, \Phi_j)_{\mathrm{TDG}}.$$

Thus, for $v = \sum_j c_j \Phi_j$, $c^* A c = A_h(v, v)$ and $c^* G c = \|v\|_{\mathrm{TDG}}^2$.

Theorem 1 (PWDG Riesz normal form)**.** *Assume that G is positive definite on the discrete Trefftz space. Then*

$$G = \frac{A - A^*}{2i}. \tag{13}$$

With $B = G^{-1/2}$, the unique Hermitian positive-definite inverse square root of G,

$$B^* A B = K + iI, \qquad K = K^*. \tag{14}$$

Hence the Riesz-normalized PWDG matrix is normal and its spectrum lies on $\operatorname{Im} z = 1$.

Proof. Let $q^\pm = \partial_{n^\pm} v^\pm$. Direct expansion on each interior edge gives

$$\operatorname{Im}(\{\{v\}\}\, [\![\nabla \bar{v}]\!]_N - \{\{\nabla v\}\} \cdot [\![\bar{v}]\!]_N) = \operatorname{Im}(v^+ \overline{q^+} + v^- \overline{q^-}).$$

After adding the boundary consistency terms and summing over elements, their total imaginary part is zero by Green's identity, because $\int_{\partial K} v \overline{\partial_{n_K} v}\, ds = \int_K (|\nabla v|^2 - k^2 |v|^2)\, dx \in \mathbb{R}$. The penalty terms give $\|v\|_{\mathrm{TDG}}^2$, so $\operatorname{Im} A_h(v, v) = \|v\|_{\mathrm{TDG}}^2$ with the positive sign in (3); polarization yields (13). Since $G > 0$, $B = G^{-1/2} = B^*$, $B^* G B = I$, and with $H = (A + A^*)/2$,

$$B^* A B = B^* H B + iI = K + iI,$$

where $K = B^* H B$ is Hermitian. $\square$

If $Hx = \mu Gx$, the μ's are invariant under nonsingular Trefftz basis changes; the normalized eigenvalues are $\mu + i$ and singular values $\sqrt{1+\mu^2}$. Thus raw $\kappa_2(A)$ is representation dependent, whereas the Riesz-normalized spectrum is intrinsic to the discrete form. If $G = V\Lambda V^*$, retaining $\lambda_j \geq \tau\lambda_{\max}$ and setting $B_\tau = V_r\Lambda_r^{-1/2}$ gives $B_\tau^* A B_\tau = K_\tau + iI_r$ on the compressed space.

The corresponding statement for the discrete UWVF is complementary and does not require exact invariance of the trace space under the continuous scattering maps. After orthonormalizing the discrete impedance-trace metric, write

$$A_{\mathrm{U}} = I - U_h, \tag{15}$$

where U_h is the discrete propagation-and-edge-exchange operator. For the absorbing problem (3), and also for an orthogonal compression of a lossless scattering operator, U_h is contractive. Therefore

$$G_{\mathrm{U}} := A_{\mathrm{U}} + A_{\mathrm{U}}^* = (I - U_h)^*(I - U_h) + (I - U_h^* U_h) \geq 0. \tag{16}$$

Whenever $G_{\mathrm{U}} > 0$, $B_{\mathrm{U}} = G_{\mathrm{U}}^{-1/2}$ gives

$$B_{\mathrm{U}}^* A_{\mathrm{U}} B_{\mathrm{U}} = \frac{1}{2}I + iK_{\mathrm{U}}, \qquad K_{\mathrm{U}} = K_{\mathrm{U}}^*. \tag{17}$$

The first term in (16) measures transmission mismatch and the second measures dissipation caused by the absorbing boundary or by discrete compression. In the lossless invariant case U_h is unitary, the second term vanishes, and $U_h q = e^{i\theta} q$ implies $G_{\mathrm{U}} q = 4\sin^2(\theta/2) q$.

5. A small numerical check

On the unit square split into two triangles, take $k = 4$ and p equispaced plane waves per element; 80-point Gauss–Legendre quadrature is used on each edge. For the conditioning table PWDG deliberately uses $\alpha = 1$, $\beta = 2$, $\delta = 1/4$, while UWVF uses its standard impedance formulation. Its diagonal blocks are outgoing-trace Gram matrices and the off-diagonal blocks pair neighboring outgoing traces with incoming test traces. Set

$$\widehat{A}_{\mathrm{P}} = G_{\mathrm{P}}^{-1/2} A_{\mathrm{P}} G_{\mathrm{P}}^{-1/2}, \qquad \widehat{A}_{\mathrm{U}} = G_{\mathrm{U}}^{-1/2} A_{\mathrm{U}} G_{\mathrm{U}}^{-1/2}.$$

As a separate check, reassembling PWDG with $\alpha = \beta = \delta = 1/2$ gives, over the same four rows,

Table 1: Raw and Riesz-normalized condition numbers.

p	$\kappa_2(A_{\mathrm{P}})$	$\kappa_2(\widehat{A}_{\mathrm{P}})$	$\kappa_2(A_{\mathrm{U}})$	$\kappa_2(\widehat{A}_{\mathrm{U}})$
5	1.23×10^1	1.225	4.18	1.192
9	3.33×10^2	1.264	1.81×10^2	1.374
13	2.48×10^5	1.392	9.12×10^4	1.755
17	7.73×10^8	1.634	3.05×10^8	2.379

$$\frac{\|A_{\mathrm{U}} + iA_{\mathrm{P}}^{(1/2)}\|_F}{\|A_{\mathrm{U}}\|_F} \leq 4.5 \times 10^{-16}, \qquad \frac{\|G_{\mathrm{U}} - 2G_{\mathrm{P}}^{(1/2)}\|_F}{\|G_{\mathrm{U}}\|_F} \leq 4.2 \times 10^{-16}.$$

Thus the classical equivalence is verified to roundoff, while Table 1 shows that the Riesz normalization remains effective for a PWDG choice not tied to the UWVF parameters.

6. Discussion

Proposition 1 and (12) explain all three classical values $\alpha = \beta = \delta = 1/2$: the separate value and normal-flux channels of PWDG become, up to the fixed factor $1/2$, an isometric representation of the oriented impedance defects used by UWVF. The methods therefore describe the same edge-coupling problem in broken-field and impedance-trace coordinates.

The Riesz normal forms sharpen this interpretation. In PWDG coordinates the skew-Hermitian part is iG_{P} and gives $K + iI$; in discrete UWVF coordinates the Hermitian defect metric $G_{\mathrm{U}} = A_{\mathrm{U}} + A_{\mathrm{U}}^*$ gives $\frac{1}{2}I + iK_{\mathrm{U}}$. This is related to operator preconditioning [4] and local UWVF boundary orthonormalization [5], but the PWDG Riesz matrix here is assembled directly from the jump and boundary terms already present in the TDG stability norm.

Under a nonsingular basis change S, $A \mapsto S^*AS$ and $G \mapsto S^*GS$: raw conditioning changes, whereas the pencil (H, G) and Riesz-normalized spectrum do not. Truncating tiny eigenvalues of G therefore separates nearly invisible Trefftz combinations from intrinsic conditioning of the discrete operator.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT (OpenAI) for assistance with language editing, LaTeX preparation, and computational verification of algebraic identities and numerical examples. The author reviewed and edited the output as needed and takes full responsibility for the content of the published article.

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